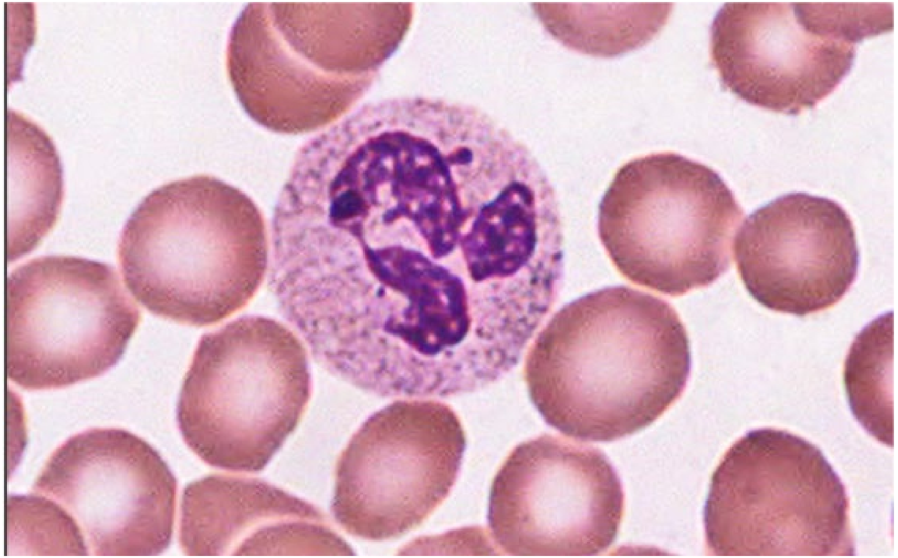
Blood and Bone Marrow Histopathology Lab

Created by Abby Cornell and MaKenna Otten

All images are from VCU Digital Histology

Neutrophil

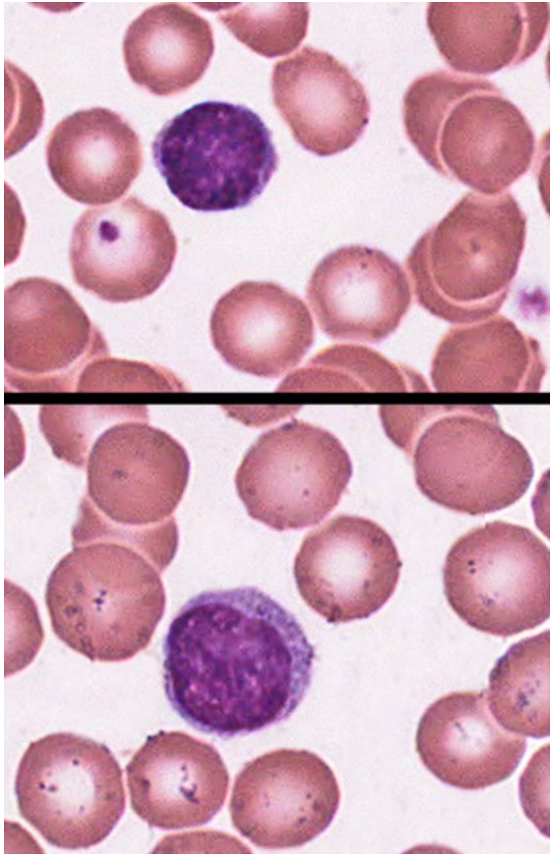


-Most abundant granulocyte

-Phagocytosis

-Lobulated nucleus

Lymphocyte



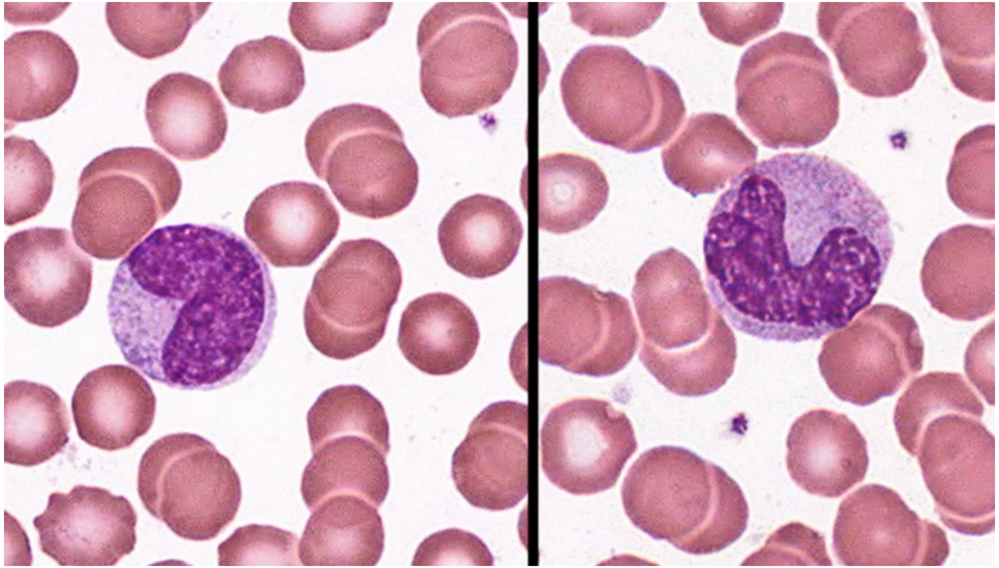
-Spherical nucleus

-Very little cytoplasm

-Cannot differentiate between T and B cells without special stain

-Agranular

Monocyte

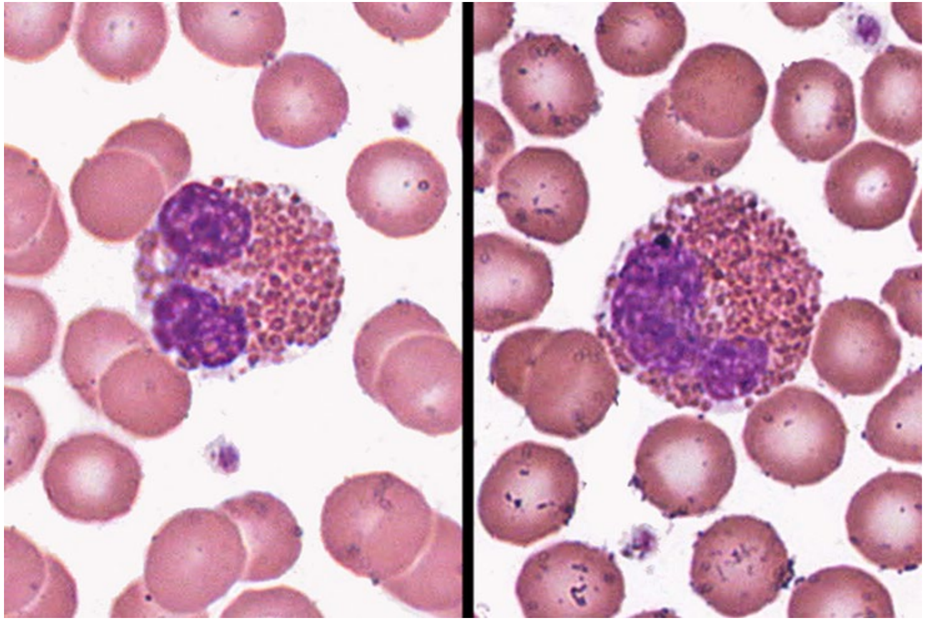


-Large, horseshoe-shaped nucleus

-Become active after entering connective tissues, where they differentiate into phagocytic cells

-Agranular

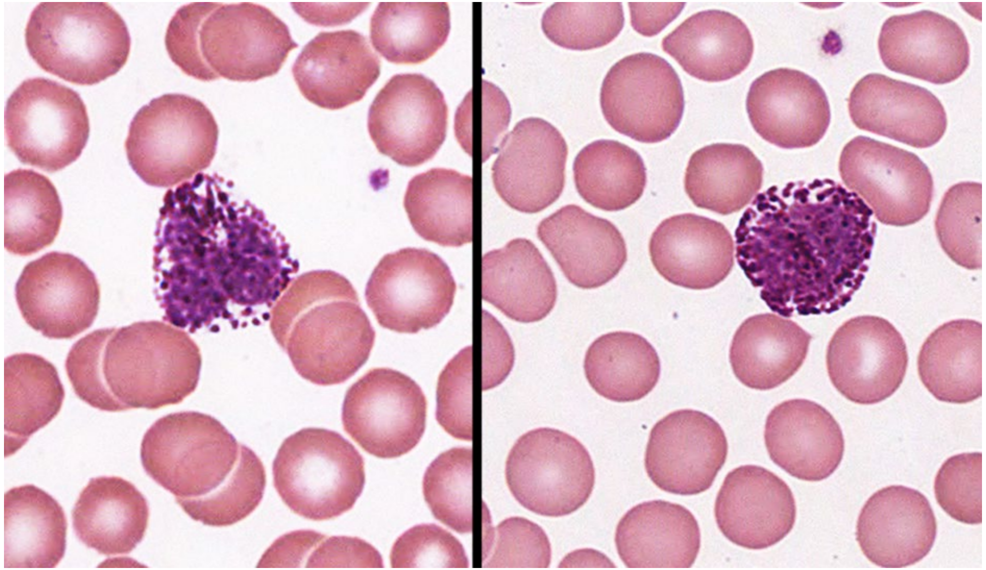
Eosinophil



-Bright eosinophilic granules in cytoplasm

-Allergic reactions, parasitic infections, phagocytosis

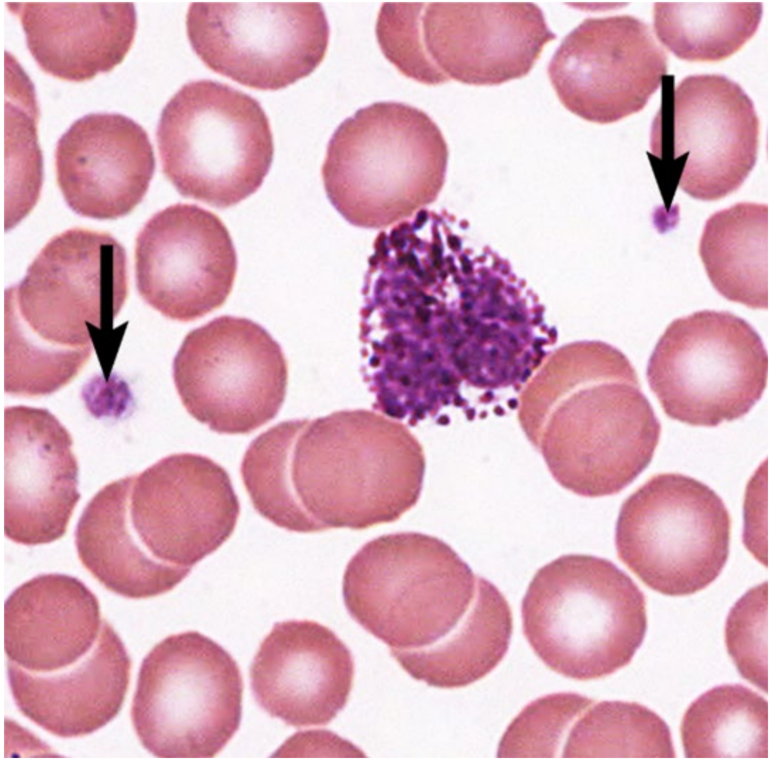
Basophil



-Large basophilic granules in cytoplasm

-Least numerous granulocyte

Platelets

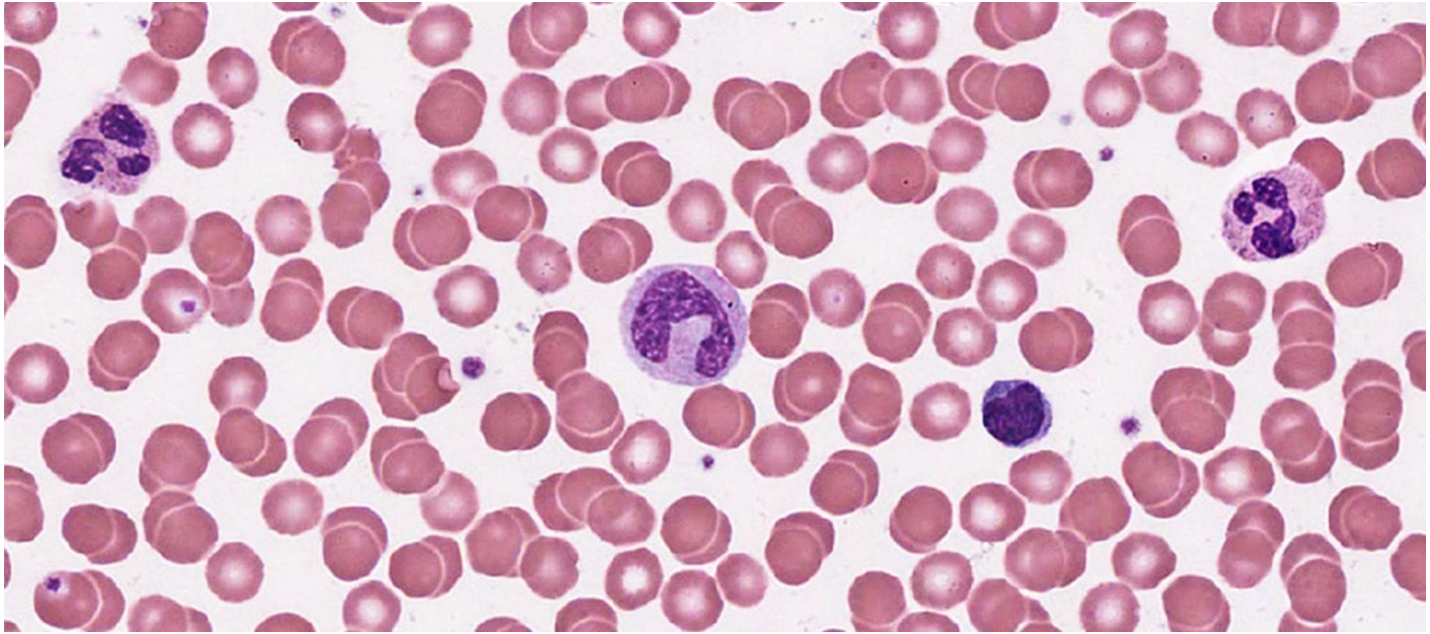


-Small, fragmented cells located in peripheral blood

-Fragments of megakaryocytes, which are found in bone marrow

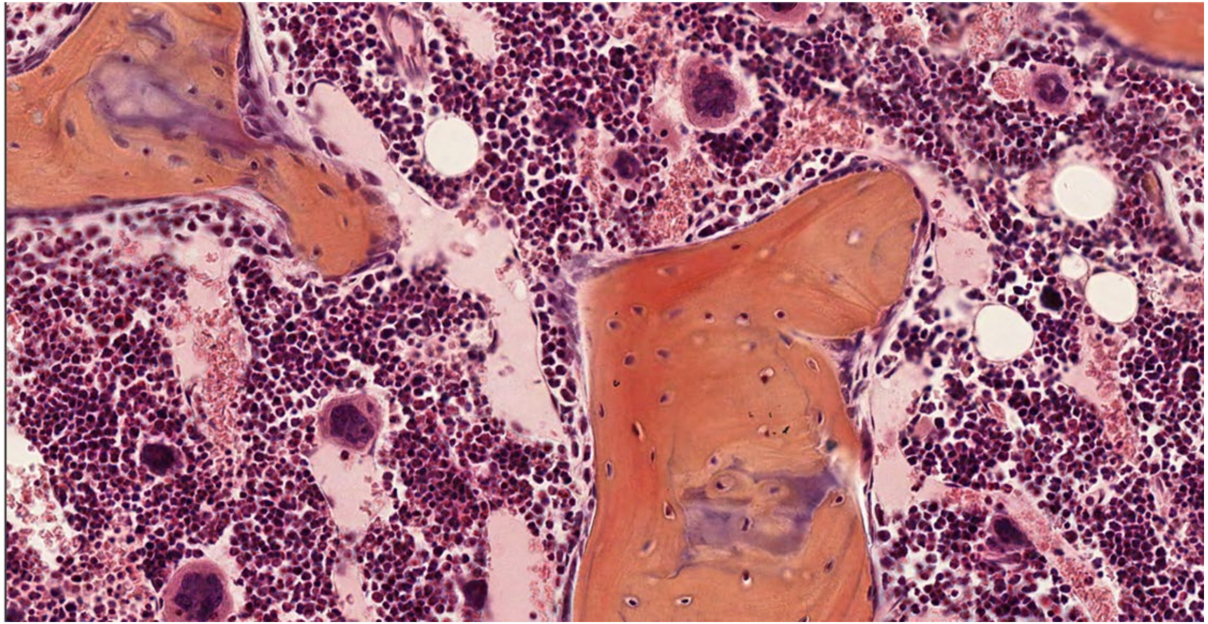
-Important functions in blood clotting and wound healing

Peripheral Blood



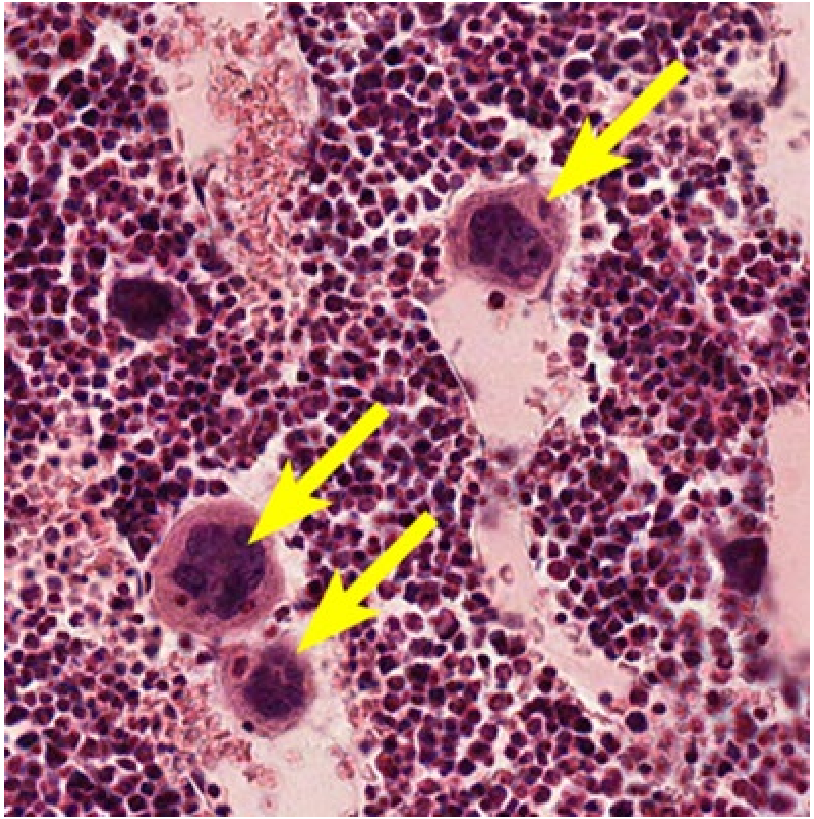
Erythrocytes, Leukocytes, Platelets

Bone Marrow



Denser and darker than peripheral blood; Contain precursor cells such as megakaryocytes

Megakaryocytes



-Precursors to platelets

-Large, multinucleated cells found in bone marrow